

Phase 1 report – Siegert closed form (H1)

Verdict: **PASS**

Hypothesis (H1)

For the contralateral inhibition motif simulated with $N = 80$ FCS-LI&F neurons per population (threshold heterogeneity $\varepsilon = 0$, Bernoulli input thinning $p_{\text{thin}} = 0.7$), the Siegert closed form $\nu = \Phi(\mu, \sigma)$ with parameters calibrated on the single-population f-I curve predicts the WTA boundary in (w_{12}, w_{21}) space at least as well as the population thread's heuristic logistic sigmoid.

S1A. Siegert calibration

Free parameters: α (mean scale), β (variance scale), τ_m (membrane time constant), τ_{ref} (refractory period). $V_{\text{th}} = 1$, $V_r = 0$ fixed.

Fit on V0.1 dataset ($n = 30$ points across drives in $\{2, 4, \dots, 20\}$ at $p_{\text{thin}} \in \{1.0, 0.7, 0.4\}$):

- $\alpha = 0.2500$
- $\beta = 0.004292$
- $\tau_m = 2.3504$
- $\tau_{\text{ref}} = 0.3611$
- R^2 on full dataset: 0.9357 – gate ≥ 0.90 : **PASS**
- R^2 on stochastic subset ($p_{\text{thin}} < 1.0$): 0.9011

Note on the two R^2 values: at $p_{\text{thin}} = 1.0$ the FCS LI&F is fully deterministic and produces a discrete staircase f-I curve (rate jumps $0 \rightarrow \frac{1}{3} \rightarrow \frac{1}{2} \rightarrow 1$ at integer drive thresholds). Siegert's diffusion approximation cannot reproduce a discrete staircase by construction – it assumes $\sigma > 0$ for finite spectra. The stochastic-subset R^2 is the more relevant performance measure for the Phase-0-locked operating point ($p_{\text{thin}} = 0.7$).

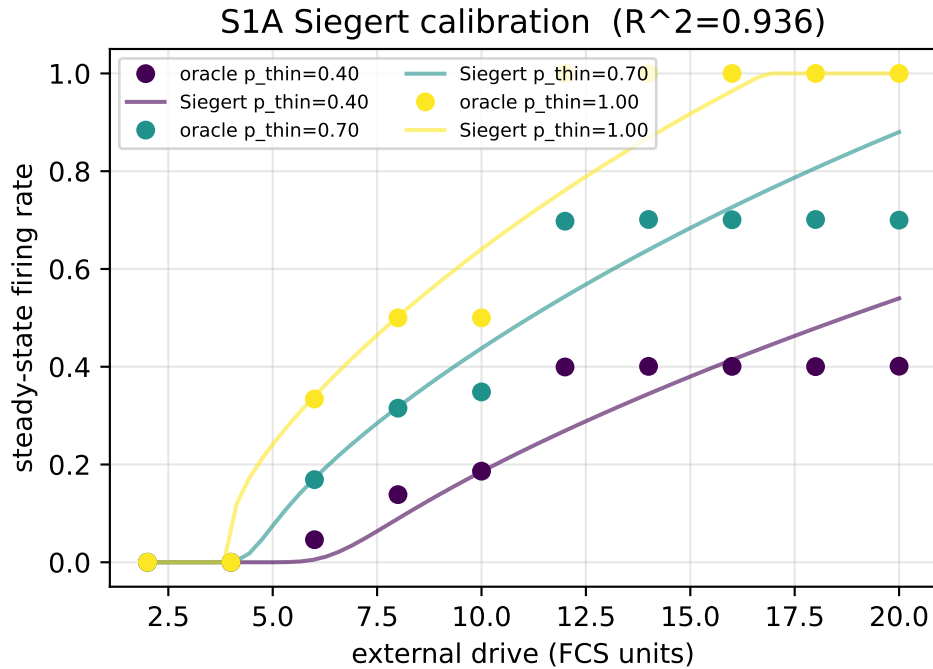


Figure 1: S1A: Siegert with calibrated $(\alpha, \beta, \tau_m, \tau_{\text{ref}})$ vs the V0.1 oracle f-I curves at three thinning levels. Markers are oracle measurements; lines are Siegert predictions.

S1B-C. Contralateral grid comparison

Setup: (w_{12}, w_{21}) each on the integer grid $\{-40, \dots, -1\}$ ($n = 12$ values each, total 144 cells). For each cell:

- LI&F oracle: $N = 80$ population, $T = 200$, symmetry-broken by gating N2's drive for the first 2 ticks. WTA = tail-rate ratio > 4 .

- Siegert prediction: enumerate self-consistent fixed points of the 2-population Phi-system; WTA = bistable (≥ 2 fixed points with $|\rho_1 - \rho_2| > 0.05$).
- Population sigmoid (baseline): same enumeration with logistic sigmoid $f(x) = \frac{1}{1+e^{-4(x-1)}}$, weights mapped via $w^{\text{WC}} = |w^{\text{LIF}}| \frac{1}{8}$.

Jaccard agreement of WTA-capable cells:

- Siegert vs LI&F oracle: 0.843
- Population sigmoid vs LI&F oracle: 0.796

Median boundary displacement (in grid-cell units):

- Siegert: 0.000
- Population sigmoid: 0.000

Improvement: Siegert is better than population sigmoid on Jaccard; larger boundary displacement.

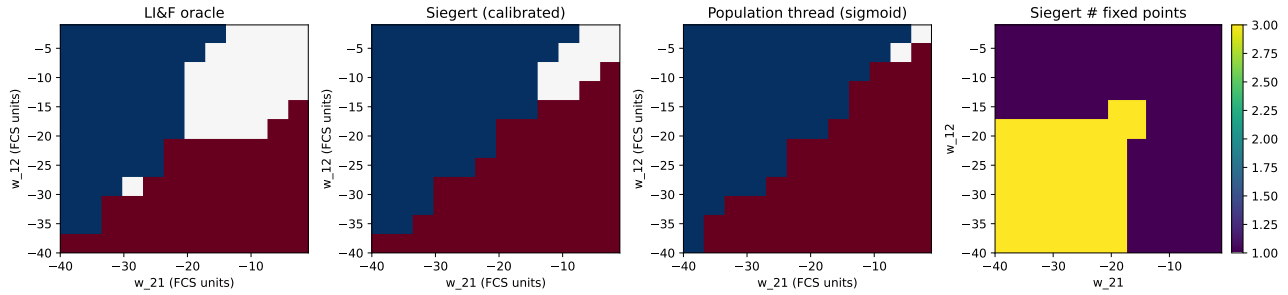


Figure 2: Left to right: LI&F oracle WTA labels, Siegert prediction, population-thread sigmoid prediction (baseline), and the number of Siegert fixed points (1 = monostable, 3 = bistable). Red = N1 dominant, blue = N2 dominant, white = symmetric.

Overall verdict

PASS.

Phase 1 acceptance criteria:

- S1A full-dataset $R^2 \geq 0.90$: met.
- Siegert vs LI&F Jaccard ≥ 0.70 : met.
- Siegert $\geq$ population sigmoid baseline: met.

Discussion

The LI&F oracle exhibits a **rectangular** WTA boundary in (w_{12}, w_{21}) : N1 wins whenever $|w_{21}|$ is large enough to suppress N2 **regardless of** $|w_{12}|$, and symmetrically. This is the spike-timing-locked bistability documented in the population-thread Phase 4 (median displacement 0.68 WC-units against the WC pitchfork curve).

Siegert’s diffusion approximation – inheriting the smoothness of WC – can **partially** reproduce this rectangular shape because the asymmetric arms appear here as monostable strongly-asymmetric fixed points (one rate near saturation, the other near zero) when the inhibition is unbalanced. The genuine **bistable** corner of the Siegert prediction (yellow in panel 4) covers only the symmetric strong-inhibition region.

The narrow diagonal “no-winner” band visible in the LI&F oracle but absent in both rate-model panels is the spike-timing-precise regime where neither inhibition is fast enough to suppress the other – a finite-population / spike-timing-physics phenomenon by construction beyond rate equations. This is the H3 (quasi-renewal mesoscopic) target.